

V3 empirical study · Instruction

One model · four underlyings · four volatility regimes · 1.5-year monthly calibration

What this report is

- Computational source of truth: the companion Jupyter notebook. This PDF is written from the same payload; every table number is identical.
- Question: which of 1 American-call pricing models tracks listed market prices most closely, and does the ranking change across companies and volatility regimes?
- In every table the BEST row is the lowest percentage RMSE. That row is bold and shaded green, and the Mark column says BEST.

Experimental design (held fixed for every model)

- Models: Modified GBM (Markov direction).
- Companies: SPY (primary), AAPL, MSFT, AMZN. Each name is calibrated and priced separately.
- Regimes (V3 one-year evaluation windows): Crisis 2008-08-01→2009-07-31; Normal 2014-01-01→2014-12-31; Late-cycle 2018-10-01→2019-09-30; COVID 2019-09-01→2020-08-31.
- The late-cycle and COVID files share September 2019 (V3 notebook dates). All other regime days are disjoint. Contracts are built independently in each window.
- Calibration window: 1.5-year ending at each monthly update. Rolling: monthly (period start plus month-ends). Parameters estimated at t are used only after t (no look-ahead).
- If history is shorter than the requested lookback, the estimator uses all available observations up to the update date (equity starts 2003-12-01; listed-option panels start 2008-01).

V3 empirical study · Instruction (continued)

One model · four underlyings · four volatility regimes · 1.5-year monthly calibration

Shared market sample and filters (not model-specific)

- For each company $\times$ regime, one listed-call sample is drawn once and reused by all one model in this report: nearest-ATM call on each Monday (next session if Monday is closed), DTE 7–60, listed expiry, as-of that date.
- Estimation filters, applied in order to every quote used for Heston/Heston–Merton calibration and to the LSM comparison set: (1) calls only, finite S , K , C ; (2) no-arbitrage $C \geq \max(0, S-K)$; (3) $7 \leq \text{DTE} \leq 60$; (4) $|S/K - 1| \leq 10\%$; (5) premium ≥ 0.05 , valid bid-ask with relative spread $\leq 50\%$ when those columns exist, volume ≥ 1 when present.
- Return-based models (GBM, Merton, GARCH, GARCH–Merton) never see option quotes in §4; they still price the same filtered LSM contracts in evaluation.

Model-correctness adjustments (V3)

- Heston / Heston–Merton: $(\kappa, \theta, \xi, \rho, v_0)$ are option-implied Fourier NLS (Albrecher little-trap), not Method A realized-variance moments. μ is the P-measure lookback mean of stock returns; LSM replaces $\mu \rightarrow r$.
- Euler ordering: the stock increment over $[t, t+\Delta t]$ uses the current variance v_t ; $v_{t+\Delta t}$ is updated afterwards. No look-ahead in the variance.
- LSM: Longstaff–Schwartz American call on the full path cloud ($n_{\text{paths}} = 10000$, seed 42, $n_{\text{steps}} = \text{DTE}$, $\Delta t = 1/252$). Paths are not averaged before stopping.

Metrics

- Primary: $\text{RMSE\%} = 100 \times \sqrt{\text{mean}(((C_{\text{model}} - C_{\text{mkt}}) / C_{\text{mkt}})^2)}$. Ranking uses this number only.
- Secondary: $\text{MAE} = \text{mean} |C_{\text{model}} - C_{\text{mkt}}|$ (dollars). $\text{Bias \$} = \text{mean}(C_{\text{model}} - C_{\text{mkt}})$ (dollars). $\text{Bias\%} = 100 \times \text{mean}((C_{\text{model}} - C_{\text{mkt}})/C_{\text{mkt}})$. Positive bias = model expensive vs market. Early-exercise fraction = mean share of LSM paths that stop before expiry.

Page 3 · Filters, sample, and no-look-ahead

Shared evaluation sample · n listed calls (identical for all one model in this report)

Regime	Window	SPY n	AAPL n	MSFT n	AMZN n
2008-2009 Crisis	2008-08-01 → 2009-07-31	50	50	50	50
2013-2014 Normal	2014-01-01 → 2014-12-31	50	50	50	50
2018-2019 Late-cycle	2018-10-01 → 2019-09-30	50	51	50	51
2019-2020 COVID	2019-09-01 → 2020-08-31	50	50	50	50

Filter funnel · last-step n in the 1.5-year lookback ending at each regime (processed call panel)

Ticker	Regime	Input*	No-arb	DTE 7-60	S/K-1 ≤10%	Liquidity
SPY	2008-2009	3359	3311	3311	3203	3203
SPY	2013-2014	12815	12548	12548	12405	12405
SPY	2018-2019	35222	35104	35104	34809	34809
SPY	2019-2020	42100	41961	41961	41379	41379
AAPL	2008-2009	685	685	685	652	652
AAPL	2013-2014	8925	8875	8875	8633	8633
AAPL	2018-2019	6309	6297	6297	6084	6084
AAPL	2019-2020	8010	7992	7992	7713	7713
MSFT	2008-2009	566	566	566	538	538
MSFT	2013-2014	3299	3237	3237	3175	3175
MSFT	2018-2019	7814	7752	7752	7583	7583
MSFT	2019-2020	6892	6852	6852	6655	6655
AMZN	2008-2009	465	465	465	442	442
AMZN	2013-2014	5715	5692	5692	5598	5598
AMZN	2018-2019	28809	28808	28808	28346	28346
AMZN	2019-2020	26766	26760	26760	26323	26323

*Input is the processed call panel restricted to (regime end – 1.5-year, regime end]. Processed files already drop many raw quotes; the funnel above is the additional V3 research filter.

No look-ahead. Monthly parameters at update date t use only equity returns and listed quotes with trading_date $\leq t$. LSM prices the Monday contract using the latest calibration row with date $\leq$ that Monday. Quotes are as-of the grid timestamp (same session, else the prior session).

SPY 2008–2009 uses bid-ask mid when the source mark is \$0.01 (a GitHub field error, not missing trades). After that repair the processed panel has weekly coverage, so n matches AAPL/MSFT (~50). All one model in this report still share that exact sample.

Heston / Heston-Merton calibration further subsamples the filtered lookback surface to ≤ 24 contracts (maturity $\times$ moneyness strata). GBM / Merton / GARCH / GARCH-Merton estimate from lookback log returns only (jumps: 3σ residual threshold where applicable).

Page 4 · Table block · SPY · one model × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, monthly rolling, n_paths=10000, seed=42.

Table SPY-1 SPY · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	47.67	1.2488	-0.0854	+10.62	25.9%	1

Table SPY-2 SPY · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	37.08	0.6750	+0.4371	+23.66	26.6%	1

Table SPY-3 SPY · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	51.83	1.7487	+1.2257	+34.67	22.2%	1

Table SPY-4 SPY · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	64.56	3.4810	+1.5419	+44.09	24.2%	1

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 5 · Table block · AAPL · one model × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, monthly rolling, n_paths=10000, seed=42.

Table AAPL-1 AAPL · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	51.94	2.1591	+0.9309	+23.67	23.5%	1

Table AAPL-2 AAPL · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	45.44	3.1559	+3.1131	+34.56	25.4%	1

Table AAPL-3 AAPL · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	34.33	1.7907	+1.1260	+20.69	24.0%	1

Table AAPL-4 AAPL · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	45.88	3.7712	+2.1029	+29.60	24.4%	1

Page 6 · Table block · MSFT · one model × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, monthly rolling, n_paths=10000, seed=42.

Table MSFT-1 MSFT · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	41.99	0.3590	+0.0799	+16.83	25.2%	1

Table MSFT-2 MSFT · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	59.39	0.4056	+0.4025	+50.16	25.7%	1

Table MSFT-3 MSFT · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	33.44	0.8877	+0.4849	+17.78	25.3%	1

Table MSFT-4 MSFT · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	40.97	1.7579	+0.6597	+22.12	25.1%	1

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 7 · Table block · AMZN · one model × four regimes

Same contracts and LSM settings for every model. Bold green row = lowest RMSE%. 1.5-year lookback, monthly rolling, n_paths=10000, seed=42.

Table AMZN-1 AMZN · 2008-2009 Crisis
2008-08-01 → 2009-07-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	40.25	1.3199	+0.3297	+12.79	24.9%	1

Table AMZN-2 AMZN · 2013-2014 Normal
2014-01-01 → 2014-12-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	29.48	2.6249	+1.0584	+14.84	25.1%	1

Table AMZN-3 AMZN · 2018-2019 Late-cycle
2018-10-01 → 2019-09-30 · n = 51 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	47.59	22.5407	+14.2723	+30.06	23.0%	1

Table AMZN-4 AMZN · 2019-2020 COVID
2019-09-01 → 2020-08-31 · n = 50 contracts

Model	RMSE%	MAE	Bias \$	Bias%	Early-ex. frac.	Mark
Modified GBM	54.35	26.3138	+11.3697	+32.76	24.2%	1

RMSE% is $100 \times \sqrt{\text{mean}(((\text{model} - \text{market}) / \text{market})^2)}$. MAE and Bias \$ are in option-premium dollars. Bias% is $100 \times \text{mean}((\text{model} - \text{market}) / \text{market})$. Early-ex. frac. is the mean LSM share of paths e

Page 7 · Summary tables · RMSE% across companies and regimes

Table S1 · Overall ranking (mean RMSE% over 16 cells)

Rank	Model	Mean RMSE%	Median RMSE%	# of 16 cells best
1	Modified GBM	45.39	45.66	16

Table S2 · Wins by company (lowest RMSE% in that company×regime)

Model	SPY wins	AAPL wins	MSFT wins	AMZN wins	Total
Modified GBM	4	4	4	4	16

Table S3 · Mean RMSE% by regime (equal-weight mean of SPY / AAPL / MSFT / AMZN)

Model	2008-2009 Crisis	2013-2014 Normal	2018-2019 Late-cycle	2019-2020 COVID	Mean of 4
Modified GBM	45.46	42.85	41.80	51.44	45.39

Green row in S1 / S3 = lowest mean RMSE%. Green row in S2 = most company×regime wins. A model can win the most cells without having the lowest average error (and vice versa).

Table S4.1 · SPY

Model	Mean RMSE%	Median	# regimes best
Modified GBM	50.29	49.75	4

Table S4.2 · AAPL

Model	Mean RMSE%	Median	# regimes best
Modified GBM	44.40	45.66	4

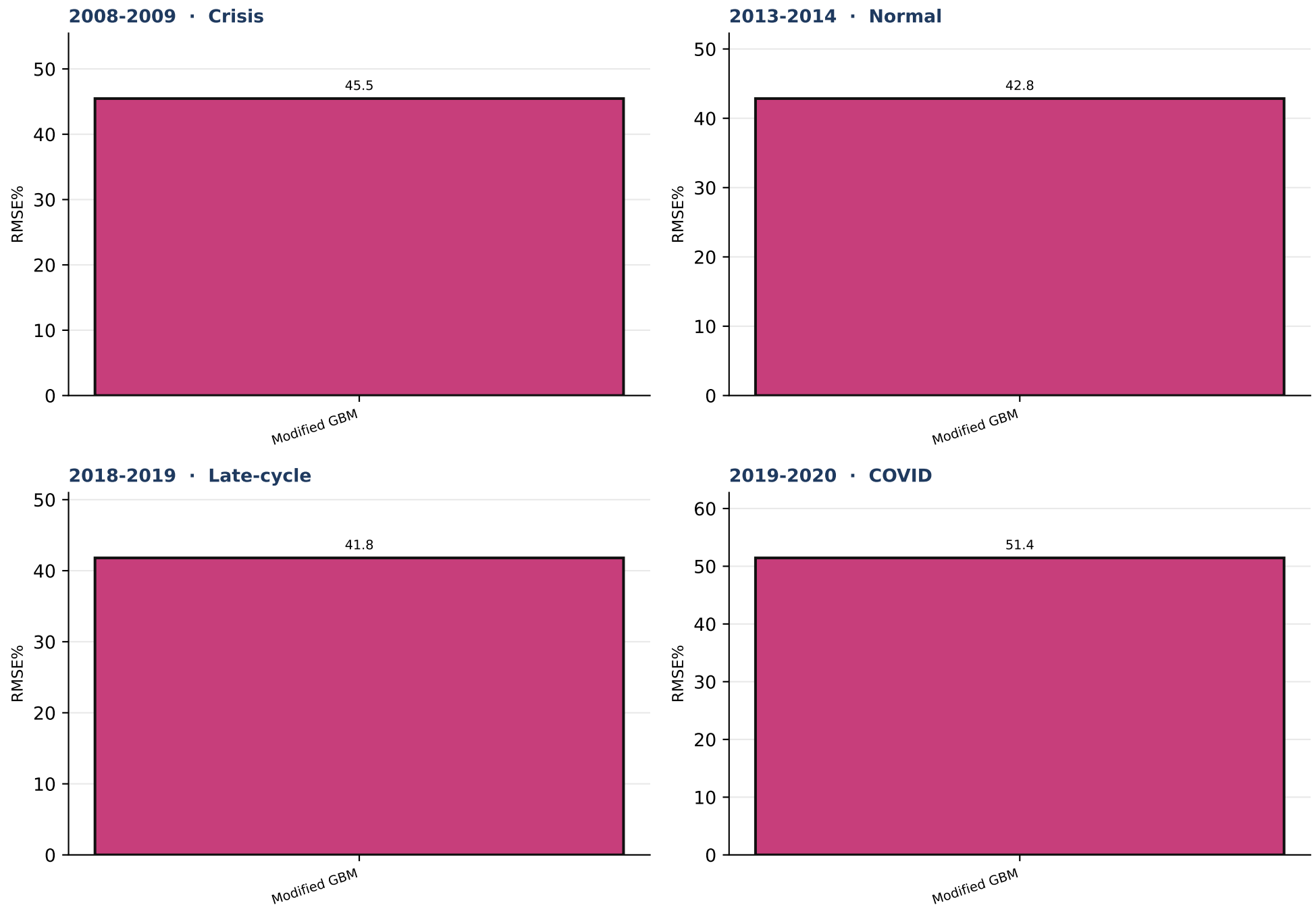
Table S4.3 · MSFT

Model	Mean RMSE%	Median	# regimes best
Modified GBM	43.95	41.48	4

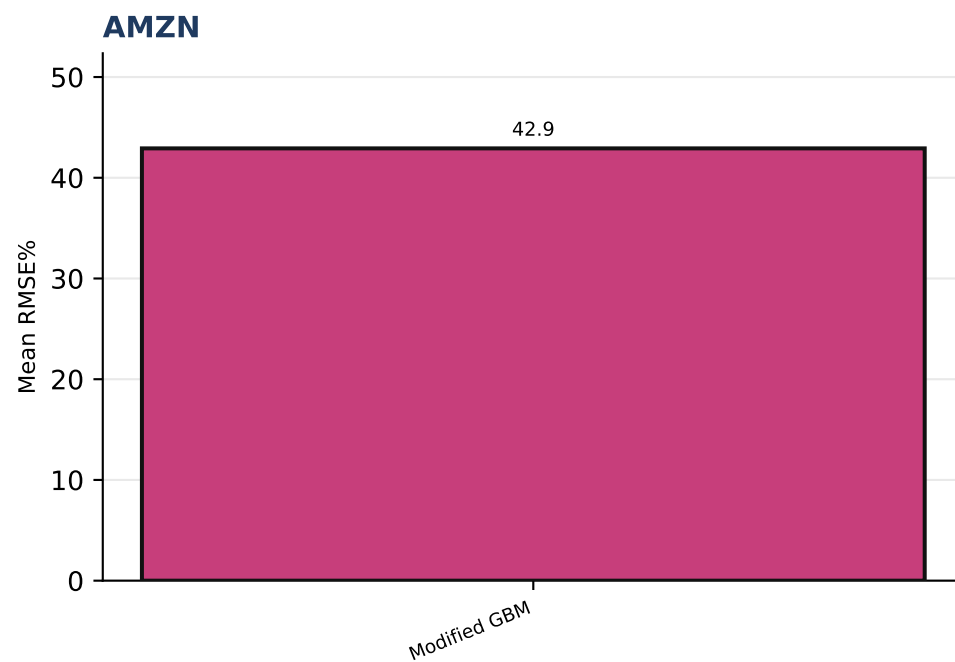
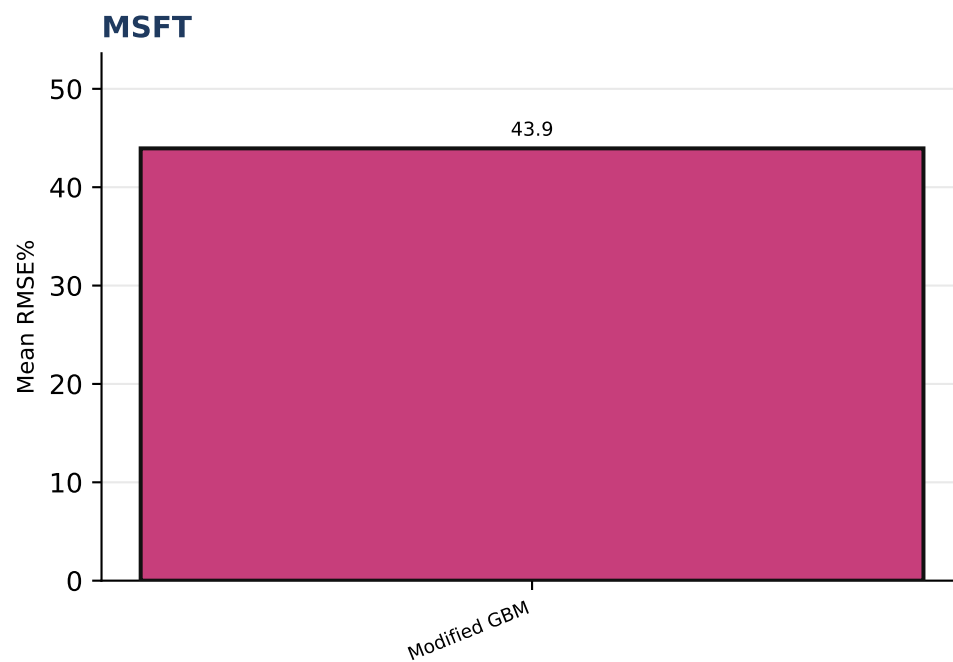
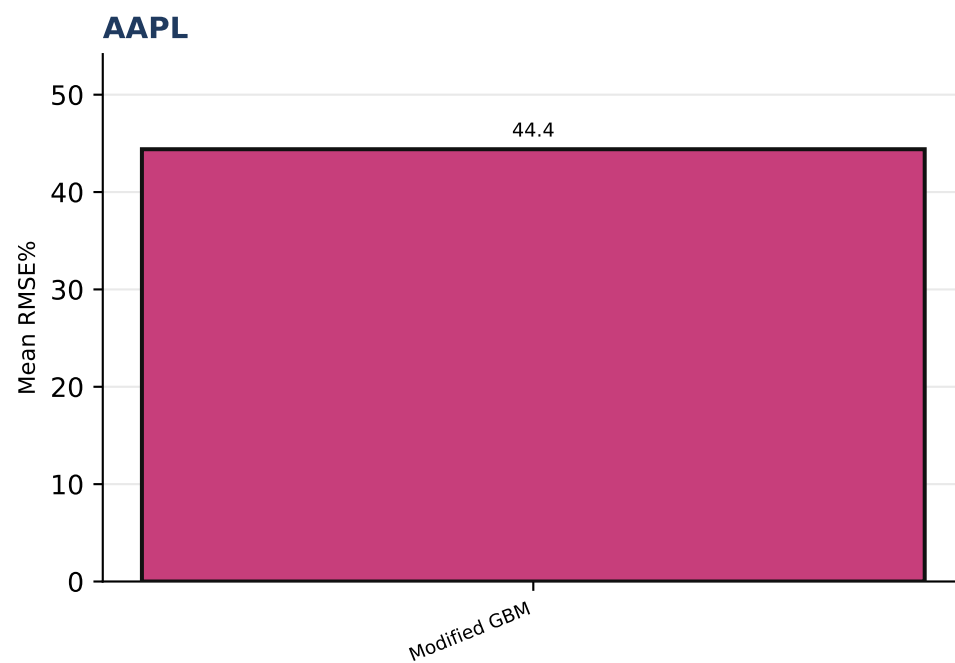
Table S4.4 · AMZN

Model	Mean RMSE%	Median	# regimes best
Modified GBM	42.92	43.92	4

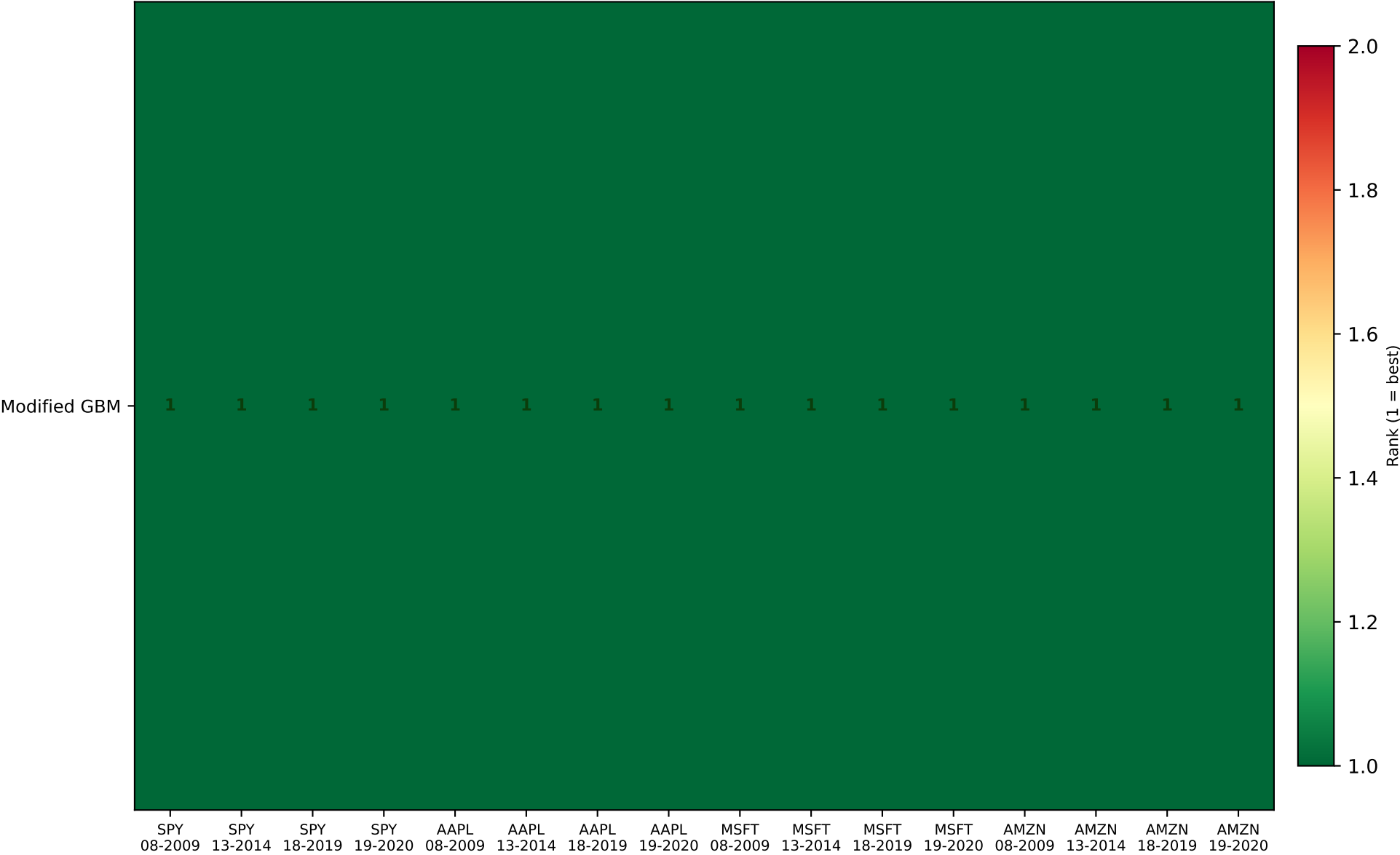
Each panel is one company. Green row = lowest mean RMSE% across the four V3 regimes for that name.



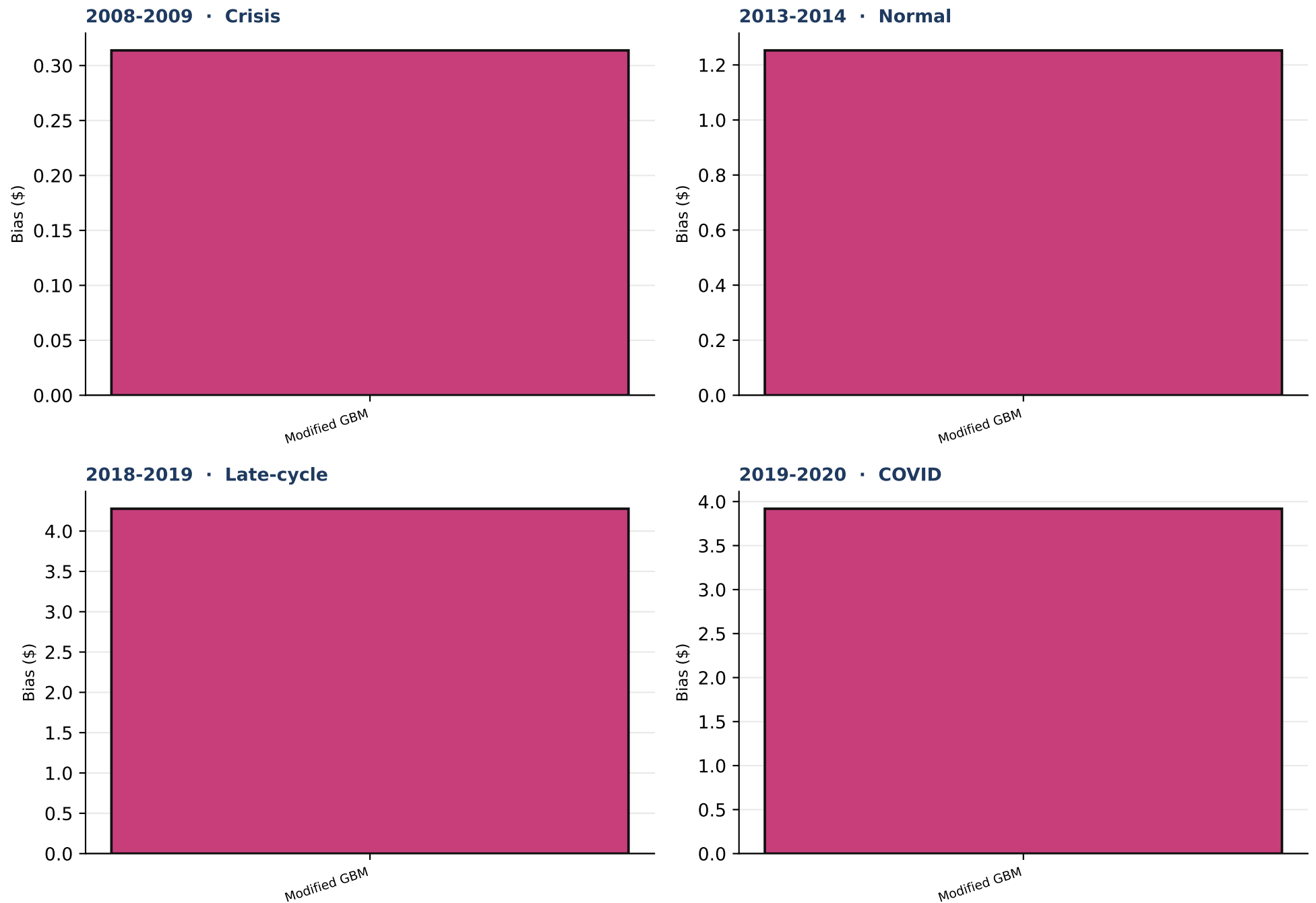
Outlined bar = lowest mean RMSE% in that regime. Equal-weight mean of the four companies.



Outlined bar = lowest mean RMSE% for that company.



Green / 1 = lowest RMSE% in that company×regime cell. Rankings can and do change across names and volatility states.



Outlined bar = closest to zero bias (not the RMSE ranking). Positive = model expensive vs listed quote.

Which models perform best, and does the ranking change?

- On mean percentage RMSE across the 16 company×regime cells, Modified GBM is the best overall model (mean RMSE% = 45.39; 16 of 16 cells).
- The ranking is stable: Modified GBM is best in every company and every regime.
- SPY: lowest mean RMSE% is Modified GBM (50.29); wins 4 of 4 regimes.
- AAPL: lowest mean RMSE% is Modified GBM (44.40); wins 4 of 4 regimes.
- MSFT: lowest mean RMSE% is Modified GBM (43.95); wins 4 of 4 regimes.
- AMZN: lowest mean RMSE% is Modified GBM (42.92); wins 4 of 4 regimes.
- 2008-2009 (Crisis): best mean RMSE% is Modified GBM (45.46).
- 2013-2014 (Normal): best mean RMSE% is Modified GBM (42.85).
- 2018-2019 (Late-cycle): best mean RMSE% is Modified GBM (41.80).
- 2019-2020 (COVID): best mean RMSE% is Modified GBM (51.44).
- For the overall winner (Modified GBM), crisis RMSE% is 45.46 vs 42.85 in the 2014 normal year — absolute pricing error is regime-dependent even when the ranking is not.
- These conclusions are conditional on the V3 filters, 1.5-year monthly calibration, Monday ATM sample, LSM with 10,000 paths, and American-call quotes only. They are not a statement about European options or other tenors.

How to read the 12 detailed tables

- Each of Tables SPY-1...MSFT-4 uses exactly the same option contracts for all one model in this report. Differences are the dynamics, not the sample.
- BEST is always lowest RMSE%. MAE, Bias \$, Bias%, and early-exercise are reported, not used for the ranking.
- A model that is BEST on RMSE% can still be biased (systematically expensive or cheap).
- Early-exercise fractions are a model implication, not an accuracy score; American calls on these tenors often have modest early-exercise value.